

# APOLLO

A Formal Specification of the Market-to-Music Pipeline:  
Spectral Analysis, Generative Sampling, and the Sonification Grammar of *TECH(NO)*  
*US*

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## Abstract

We give a complete mathematical specification of a pipeline that maps a financial price history to a musical composition. Part I establishes the second-order theory: the Wiener–Khinchin representation of the power spectral density (PSD), its estimators (periodogram, Welch, Blackman–Tukey, Thomson multitaper), Fisher’s exact test for hidden periodicities, and the Geweke–Porter–Hudak (GPH) estimator of long memory. Part II specifies three generative models admitting exact or approximate sampling from the estimated law: phase-randomized surrogates, GJR-GARCH with skew- $t$  innovations, and a recurrent conditional-density network. Part III defines the sonification grammar: a measurable map from the path of the (real or sampled) return process to a discrete musical event set and thence to a pressure waveform, under which distinct moments of the conditional distribution are carried by perceptually orthogonal audio dimensions — location by pitch, scale by loudness, and tail mass by timbre.

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\*Pipeline implementation: [github.com/vitociciretti/apollo](https://github.com/vitociciretti/apollo). This document specifies the mathematics implemented in `pipeline/`.

## 1 Data and notation

Let  $P_t > 0$ ,  $t = 0, 1, \dots, n$ , denote the dividend- and split-adjusted daily closing price of an asset on trading-day grid  $t$ . Define log-returns

$$r_t := \log P_t - \log P_{t-1}, \quad t = 1, \dots, n. \quad (1)$$

Throughout,  $\hat{\mu} := n^{-1} \sum_t r_t$ ,  $\hat{s}^2 := n^{-1} \sum_t (r_t - \hat{\mu})^2$ , and the *volatility proxy* is  $a_t := |r_t|$ . Annualization uses the convention  $\sigma_{\text{ann}} = \hat{s} \sqrt{252}$ . The cumulative log-price is  $X_t := \sum_{u \leq t} r_u$ ; the running maximum  $M_t := \max_{u \leq t} X_u$ ; and the *drawdown process*

$$D_t := M_t - X_t \geq 0. \quad (2)$$

**Assumption 1.1.**  $(r_t)$  is covariance-stationary with autocovariance  $\gamma(h) = \text{Cov}(r_t, r_{t+h})$  and  $\sum_h |\gamma(h)|$  possibly divergent (long memory is admitted for  $a_t$ ; see §2.3).

## 2 Second-order structure and the Wiener–Khinchin theorem

**Theorem 2.1** (Wiener–Khinchin). *Let  $(Y_t)_{t \in \mathbb{Z}}$  be covariance-stationary with autocovariance  $\gamma(\cdot)$ . There exists a finite measure  $F$  on  $[-1/2, 1/2]$  (the spectral measure) with*

$$\gamma(h) = \int_{-1/2}^{1/2} e^{2\pi i f h} dF(f), \quad h \in \mathbb{Z}. \quad (3)$$

If  $\sum_h |\gamma(h)| < \infty$ ,  $F$  is absolutely continuous with density

$$S(f) = \sum_{h=-\infty}^{\infty} \gamma(h) e^{-2\pi i f h}, \quad f \in [-1/2, 1/2], \quad (4)$$

the power spectral density, and  $\gamma(h) = \int e^{2\pi i f h} S(f) df$ ; in particular  $\text{Var}(Y_t) = \int_{-1/2}^{1/2} S(f) df$ .

Equation (4) is the load-bearing identity of the project: *the PSD is the Fourier transform of the autocovariance*, hence estimating  $S$  and estimating  $\gamma$  are the same problem, and any generative model matching  $S$  matches all second-order properties of the process.

### 2.1 Estimators implemented

Write  $Y_t$  for the demeaned series of interest ( $r_t - \hat{\mu}$  or  $a_t - \bar{a}$ ).

**(i) Periodogram.** With the DFT  $d(f_j) = \sum_{t=1}^n Y_t e^{-2\pi i f_j t}$  at Fourier frequencies  $f_j = j/n$ ,  $j = 1, \dots, m = \lfloor (n-1)/2 \rfloor$ ,

$$I(f_j) = \frac{1}{n} |d(f_j)|^2. \quad (5)$$

$I$  is asymptotically unbiased for  $S$  but inconsistent ( $I(f_j)/S(f_j) \xrightarrow{d} \frac{1}{2}\chi_2^2$ , variance not vanishing), so it is used only where its *exact distribution* is exploited (Fisher’s test, §2.2; GPH, §2.3).

(ii) **Welch.** Split  $Y$  into  $K$  segments of length  $L$  (implementation:  $L = 512$ , 75% overlap), taper each with a Hann window  $w$ , and average the windowed periodograms:

$$\hat{S}_W(f) = \frac{1}{K} \sum_{k=1}^K \frac{1}{\sum_t w_t^2} \left| \sum_{t=1}^L w_t Y_{t+o_k} e^{-2\pi i f t} \right|^2. \quad (6)$$

Variance decays like  $1/K$  at the cost of frequency resolution  $\asymp 1/L$ .

(iii) **Blackman–Tukey (the Wiener–Khinchin route).** Estimate  $\hat{\gamma}(h) = n^{-1} \sum_{t=1}^{n-h} Y_t Y_{t+h}$  for  $|h| \leq L$  (implementation:  $L = 512$ ), damp with the Tukey–Hanning lag window  $\lambda(h) = \frac{1}{2}(1 + \cos \frac{\pi h}{L})$ , and transform:

$$\hat{S}_{BT}(f) = \sum_{|h| \leq L} \lambda(h) \hat{\gamma}(h) e^{-2\pi i f h}. \quad (7)$$

This estimator is Theorem 2.1 made operational; agreement of  $\hat{S}_{BT}$  with  $\hat{S}_W$  is used as an internal consistency check.

(iv) **Thomson multitaper.** Let  $v^{(0)}, \dots, v^{(K-1)}$  be the discrete prolate spheroidal sequences (DPSS) of length  $n$  and time–bandwidth product  $NW$  (implementation:  $NW = 4$ ,  $K = 7$ ), the orthonormal maximizers of spectral concentration in  $[-W, W]$ . The estimator

$$\hat{S}_{MT}(f) = \frac{1}{K} \sum_{k=0}^{K-1} \left| \sum_{t=1}^n v_t^{(k)} Y_t e^{-2\pi i f t} \right|^2 \quad (8)$$

has  $\approx 2K$  equivalent degrees of freedom and minimal broadband leakage, making it the estimator of record for detecting weak spectral *lines* against a colored background — precisely the regime of volatility spectra.

## 2.2 Fisher’s exact test for hidden periodicities

Under  $H_0: r_t \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$ , the normalized ordinates  $I(f_j)/\sum_k I(f_k)$  are exchangeable and Fisher’s statistic

$$g = \frac{\max_{1 \leq j \leq m} I(f_j)}{\sum_{j=1}^m I(f_j)} \quad (9)$$

has the exact null law

$$\Pr(g > x) = \sum_{k=1}^{\lfloor 1/x \rfloor} (-1)^{k-1} \binom{m}{k} (1 - kx)^{m-1}. \quad (10)$$

A per-ordinate 5% false-alarm level (Bonferroni across  $m$  ordinates) is  $x^* = 1 - (0.05/m)^{1/(m-1)}$ .

**Empirical result (NVDA, 2010–2026,  $n = 4156$ ).** For  $r_t$ :  $g$  attains  $p = 0.34$  and zero ordinates clear  $x^*$  — the return spectrum is statistically flat (white noise), consistent with the martingale hypothesis. For  $a_t = |r_t|$ :  $p < 10^{-4}$  with eight ordinates above the false-alarm level, all at low frequency, including periods  $\approx 63$  and  $\approx 126$  trading days — the quarterly and semi-annual earnings-calendar harmonics.

**Remark 2.2.** Once long memory in  $a_t$  is established (§2.3) the white-noise null is weak for individual low-frequency peaks: power concentration under a  $1/f$ -shaped spectrum inflates (10) rejections. The robust finding is the long-memory slope plus the calendar harmonics, not any single line.

### 2.3 Long memory: fractional integration and GPH

**Definition 2.3.**  $(Y_t)$  has long memory with parameter  $d \in (0, \frac{1}{2})$  if  $S(f) \sim c f^{-2d}$  as  $f \downarrow 0$ , equivalently  $\gamma(h) \sim c' h^{2d-1}$  as  $h \rightarrow \infty$ .

The Geweke–Porter–Hudak estimator regresses log-periodogram ordinates on the canonical regressor. With  $\lambda_j = 2\pi f_j$  and bandwidth  $m^* = \lfloor n^{0.6} \rfloor$ ,

$$\log I(f_j) = c - d \log(4 \sin^2(\lambda_j/2)) + \varepsilon_j, \quad j = 1, \dots, m^*, \quad (11)$$

estimated by OLS; asymptotically  $\hat{d} \approx \mathcal{N}(d, \pi^2/(24 m^*))$ , i.e.  $\text{se}(\hat{d}) = \pi/\sqrt{24 m^*}$ .

**Empirical result.** For NVDA  $a_t$ :  $\hat{d} = 0.397$ ,  $\text{se} = 0.053$  ( $t = 7.5$ ), squarely in the FIGARCH range documented for equity volatility. For  $r_t$  the low-frequency log–log slope is  $-0.04$  (white noise = 0).

## 3 Generative models: sampling from the estimated law

The composition engine (§4) consumes an arbitrary path  $(r_t^*)_{t \leq N}$ ; three samplers of increasing structural fidelity are implemented.

### 3.1 Phase randomization (maximum entropy given the ACF)

Let  $A_j = |\sum_t (a_t - \bar{a}) e^{-2\pi i j t/n}|$  be the empirical amplitude spectrum, interpolated to target length  $N$  and rescaled by  $\sqrt{N/n}$ . Draw  $\varphi_j \stackrel{iid}{\sim} U[0, 2\pi)$ ,  $\varphi_0 = 0$ , and set

$$a_t^* = \bar{a} + \frac{1}{N} \sum_j A_j e^{i\varphi_j} e^{2\pi i j t/N}, \quad a_t^* \leftarrow \max(a_t^*, 0). \quad (12)$$

By construction  $(a_t^*)$  has (up to interpolation error) the sample autocovariance of  $(a_t)$ ; among all processes with that autocovariance the Gaussian one generated here maximizes entropy. This is Theorem 2.1 used as a *sampler*: the PSD is a complete generative model for the second-order law.

### 3.2 GJR-GARCH(1, 1, 1) with skew- $t$ innovations

$$r_t = \mu + \sigma_t z_t, \quad z_t \stackrel{iid}{\sim} \text{skew-}t(\eta, \lambda), \quad (13)$$

$$\sigma_t^2 = \omega + (\alpha + \gamma \mathbf{1}\{r_{t-1} < \mu\}) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \varepsilon_t = r_t - \mu, \quad (14)$$

estimated by maximum likelihood. The leverage term  $\gamma > 0$  generates the volatility asymmetry that the linear samplers of §3.1 cannot. Fitted on NVDA ( $\times 100$  scale):  $\hat{\mu} = 0.116$ ,  $\hat{\omega} = 0.073$ ,  $\hat{\alpha} = 0.030$ ,  $\hat{\gamma} = 0.070$ ,  $\hat{\beta} = 0.932$ ,  $\hat{\eta} = 4.57$ ,  $\hat{\lambda} = -0.02$ ; persistence  $\hat{\alpha} + \hat{\gamma}/2 + \hat{\beta} \approx 0.997$ .

### 3.3 Recurrent conditional-density model

Let  $x_t = (r_t/\hat{s}, \log(|r_t|/\hat{s} + 10^{-3})) \in \mathbb{R}^2$ . A single-layer LSTM  $h_t = \text{LSTM}(x_t, h_{t-1}) \in \mathbb{R}^{64}$  feeds an affine head producing the parameters of the *conditional law of tomorrow's standardized return*:

$$\mu_t = 0.05 \tanh(w_\mu^\top h_t), \quad \sigma_t = \exp(\text{clip}(w_\sigma^\top h_t, [-4, 2])), \quad \nu = 2.05 + \text{softplus}(\theta_\nu), \quad (15)$$

$$\frac{r_{t+1}}{\hat{s}} \Big| \mathcal{F}_t \sim \mu_t + \sigma_t T_\nu, \quad (16)$$

with  $T_\nu$  standard Student- $t$ . Training minimizes the exact negative log-likelihood

$$-\ell = -\log \Gamma\left(\frac{\nu+1}{2}\right) + \log \Gamma\left(\frac{\nu}{2}\right) + \frac{1}{2} \log(\pi\nu) + \log \sigma_t + \frac{\nu+1}{2} \log\left(1 + \frac{z_t^2}{\nu}\right), \quad z_t = \frac{r_{t+1}/\hat{s} - \mu_t}{\sigma_t}, \quad (17)$$

over overlapping windows (length 256, stride 16, 32-step burn-in excluded from the loss), Adam with gradient clipping, model selection by validation NLL on a 15% holdout. Sampling is autoregressive: the hidden state is warmed on the final 252 real observations, then  $r_{t+1}^* = \hat{s}(\mu_t + \sigma_t T_\nu)$  is drawn, clipped at 8 standard units, and fed back through  $x_{t+1}$ .

**Diagnostics (real vs. sampled,  $N = 4608$ ).**

|                        | NVDA             | LSTM sample    |
|------------------------|------------------|----------------|
| tail d.o.f.            | 4.57 (GARCH MLE) | 4.24 (learned) |
| excess kurtosis basis  | 9.8              | 10.1           |
| ACF $_{ r }(1)$        | 0.171            | 0.131          |
| ACF $_{ r }(21)$       | 0.092            | 0.089          |
| GPH $\hat{d}$ on $ r $ | 0.397            | 0.516          |
| annualized vol         | 45.2%            | 53.4%          |
| mean daily return      | +0.148%          | +0.044%        |

Fat tails, volatility clustering and long memory are recovered from raw likelihood training; the conditional mean is (correctly) shrunk toward zero, which materially darkens the sampled paths' drawdown structure.

## 4 The sonification grammar

### 4.1 Grids and state processes

Fix beats per minute  $B$  and the sixteenth-note duration  $\tau := \frac{60}{4B}$  seconds. Days are binned  $\delta$  per sixteenth,  $\delta = \max(2, \text{round}(n/1536))$ , giving  $K = 16 \lfloor n/(16\delta) \rfloor$  sixteenth-notes and  $K/16$  bars. Define on the day grid: rolling volatility  $v_t = \text{sd}(r_{t-20:t})\sqrt{252}$ ; rolling kurtosis  $\kappa_t = \widehat{\text{kurt}}(r_{t-62:t})$ ; trend z-score

$$z_t = \frac{X_t - \text{MA}_{252}(X)_t}{\text{sd}_{252}(X)_t}; \quad (18)$$

and the *regime indicator* from the drawdown (2),

$$\beta_t = \mathbf{1}\{D_t > Q_{0.78}(D)\}, \quad (19)$$

i.e. breakdown  $\equiv$  worst-quartile drawdown *of the path's own distribution* (adaptive by construction: any sampled world keeps  $\approx 22\%$  breakdown time). Bar-level states  $(z_b, v_b, \kappa_b, \beta_b)$  evaluate these at each bar's final day;  $v_b$  is min-max normalized to  $\tilde{v}_b \in [0, 1]$  and  $\kappa_b$  is mapped to its percentile rank  $\tilde{\kappa}_b \in [0, 1]$ .

### 4.2 Pitch: scale quantization

Let the key root be pitch class  $\rho$  (per track, Table 1) and octave-anchored MIDI root  $R$ . Modes are  $\mathcal{S}_{\text{aeol}} = \{0, 2, 3, 5, 7, 8, 10\}$  and  $\mathcal{S}_{\text{phry}} = \{0, 1, 3, 5, 7, 8, 10\}$ , selected by  $\beta$ :  $\mathcal{S}_b = \mathcal{S}_{\text{phry}}$  if  $\beta_b = 1$  else  $\mathcal{S}_{\text{aeol}}$ . The harmony degree per bar is

$$\text{deg}_b = \begin{cases} 5 & z_b < -1, \\ 0 & -1 \leq z_b \leq 0.5, \\ 3 & z_b > 0.5, \end{cases} \quad m_b = R + \mathcal{S}_b[\text{deg}_b]. \quad (20)$$

Pitch is thus a quantized functional of price-versus-trend; the mode (and hence the semitone content) is a functional of the drawdown regime.

### 4.3 Event maps

For sixteenth  $k$  in bar  $b$  with day window  $W_k = [k\delta, (k+1)\delta)$ :

|                      |                                                                                                                 |                                              |
|----------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>kick:</b>         | $k \bmod 4 = 0 \wedge \beta_b = 0$                                                                              | velocity $\propto 0.85 + 0.15 \tilde{v}_b$ , |
| <b>bass note:</b>    | $\beta_b = 0$ , pitch $m_b + 12 \cdot \mathbf{1}_{\{\max_{t \in W_k}  r_t  > \hat{s} \wedge k \bmod 4 = 2\}}$ , | off-beat accent $\times 1.15$ ,              |
| <b>stab (chord):</b> | $k \bmod 16 \in \{3, 7, 10, 14\} \wedge \sum_{t \in W_k}  r_t  > \delta \overline{ r }$ ,                       | triad on $\deg_b$ ,                          |
| <b>clap roll:</b>    | $\beta_b = 1 \wedge \beta_{b+1} = 0$                                                                            | velocity ramp in $k \bmod 16$ ,              |
| <b>crash:</b>        | $\max_{t \in W_k}  r_t  > 4\hat{s} \vee$ (first $k$ of a recovery bar).                                         |                                              |

The filter-cutoff automation (MIDI CC74; audio-domain lowpass corner) is

$$c_b = 160 + 2800 \tilde{v}_b + 500 \tilde{\kappa}_b \text{ Hz}, \quad (21)$$

so the audible brightness of the bass *is* the volatility curve.

### 4.4 Timbre as a tail functional

Each melodic voice is a detuned additive synth

$$y(t) = \sum_{\pm} \sum_{h=1}^H \frac{\alpha_h}{h} \sin(2\pi h f(1 \pm \epsilon)t), \quad \alpha_h = \tilde{\kappa} \mathbf{1}_{\{h \geq 2\}}, \quad \epsilon = 0.004 + 0.006 \tilde{\kappa}, \quad (22)$$

band-limited at the filter corner: at  $\tilde{\kappa} = 0$  a pure sine (thin-tailed, quasi-Gaussian regime), at  $\tilde{\kappa} = 1$  a bright sawtooth (crash-prone regime). Thus the fourth moment is carried by harmonic content — the “piano vs. violin” axis — while the second moment is carried by loudness and the first by pitch: three perceptually orthogonal channels for three moments.

### 4.5 Pads: the spectrum made audible

For each significant volatility-spectrum peak at frequency  $f_0$  (period  $1/f_0 \in [40, 600]$  days), the band-limited analytic envelope is extracted by zero-phase Butterworth bandpass  $\mathcal{B}_{[f_0/1.35, 1.35f_0]}$  and the Hilbert transform  $\mathcal{H}$ :

$$E_{f_0}(t) = \text{smooth} \left| (\text{id} + i\mathcal{H}) \mathcal{B}_{[f_0/1.35, 1.35f_0]}(a - \bar{a})(t) \right|, \quad (23)$$

i.e. the instantaneous amplitude of that volatility cycle through time. The pad voice sounds at the peak frequency transposed into the audible range,  $f_{\text{aud}} = 2.2 \times 10^5 f_0$  Hz, quantized to the track’s scale lattice  $\{440 \cdot 2^{(m-69)/12} : m \bmod 12 \in \rho + \mathcal{S}\}$ , amplitude-modulated by  $E_{f_0}$ , and gated by the smoothed regime indicator (19) so the spectrum is heard precisely when the beat is absent.

### 4.6 Mixing operators

Sidechain compression is the multiplicative envelope  $g(t) = 1 - 0.55 e^{-(t-t_{\text{kick}})/0.085}$  applied to all non-drum buses after each kick onset; spatialization applies a dotted-eighth ( $3\tau$ ) cross-feedback delay to the  $> 1.2$  kHz band; mastering is a 25 Hz DC-blocking highpass and peak normalization (no nonlinear stage — an earlier tanh soft-clipper was removed as it injected broadband intermodulation audible as background noise).

## 5 Album parameterization

Tempo is an affine function of annualized volatility,  $B = \text{clip}(\lfloor 120 + 40(\sigma_{\text{ann}} - 0.20) \rfloor, 122, 134)$ ; each asset receives a fixed key.

| #  | Title            | Key    | $\sigma_{\text{ann}}$ | BPM |
|----|------------------|--------|-----------------------|-----|
| 01 | Chipmaker        | F min  | 45%                   | 130 |
| 02 | Cupertino        | A min  | 28%                   | 123 |
| 03 | Redmond          | D min  | 26%                   | 122 |
| 04 | PageRank         | G min  | 28%                   | 123 |
| 05 | Everything Store | C min  | 33%                   | 125 |
| 06 | Social Graph     | Bb min | 40%                   | 127 |
| 07 | Stream           | E min  | 49%                   | 131 |
| 08 | Ludicrous        | B min  | 57%                   | 134 |

Table 1: *TECH(NO) US*: per-track parameters derived from each asset’s history.

## 6 Video synchronization

The video frame at time  $s \in [0, K\tau]$  displays days  $1, \dots, \lfloor sK\delta/(K\tau) \rfloor = \lfloor s\delta/\tau \rfloor$ : the map from audio time to market time is linear with slope  $\delta/\tau$  days per second ( $\approx 63$  trading days/s at  $B = 128$ ,  $\delta = 3$ ). Overlays are the regime shading  $\{\beta_t = 1\}$  (visible iff the playhead has passed the region’s onset), the running price  $e^{X_t}$  on a log axis,  $v_t$  in a lower panel, and markers at  $\{t : |r_t| > 4\hat{s}\}$  — the visual twins of the breakdown, the filter curve, and the crash cymbal respectively.

### Caveat

A flat return spectrum (§2.2) is exactly the statement that the composition carries no linear predictive content. Apollo is prophecy only in the sense that mattered at Delphi: accurate, resonant, and of no use for trading.